# Dr Alexandros Filotheou


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|---|---|
| Current location and age | Leuven, Belgium \| 39 |
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I have 8+ years of hands-on experience in robotics which includes SLAM, Localisation & Pose Estimation, Autonomous Navigation, Control, Computer Vision, task planning with Behaviour Trees, and general integration, problem-solving, and troubleshooting. These I have acquired in real conditions with real robots as well as in simulation, working for the industry, for research projects funded by the European Commission and the Greek State, or through volunteering. I am proficient in ROS and ROS 2, but also MATLAB/Octave, all the while under Linux, which has been my primary OS since 2008. My primary coding language is C++ and my secondary Python.

I hold a diploma and a PhD in Electrical and Computer Engineering and I am qualified with a Master's degree in Control and Robotics. I have deliberately chosen the academic + software engineering path because I wanted to reconcile the two dominant types of engineers: the academic type whose coding boundary is MATLAB, and the software engineering type whose theoretic boundary remains in code.

It is my pleasure to document my motivation and contributions in a clear and concise manner that acknowledges my audience's existing knowledge with regard to code documentation, presentations, technical papers, or tutorials. Most of all I enjoy being a member of a team and solving problems together with other people.

## Quick Links

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|---|---|
| Compact version of this CV | On `github` |
| Indicative software packages: | `cbgl` · `fsm-lo` · `lama-odom` · `pvhd` — `github`<br>`hitch_estimation_apriltag_array` · `pointcloud_conversions` · `gitrl-z` |
| Demos / videos: | Integration [1] · [2] · [3] · SLAM/Navigation<br>`cbgl` · `cultureid` · `fsm` · `relief` · `mpc` |
| Indicative publications: | [1] [2] [3] [4] — google scholar |
| Portfolio | On `github` |
| References | On `github` |

## Contents


1 **Work Experience**
2 **Experience in R&D Projects**
3 **Voluntary Experience**
4 **Open Source Contributions**
5 **Education**
6 **Publications**
7 **Distinctions**
8 **Computer Skills**
9 **Languages**
10 **References**

## Work Experience

2026.02 – present **Senior Robotics Software Engineer**
Intermodalics BV, Leuven, Belgium
Bridging of a VDA5050 fleet manager with ROS 2 mission behaviours, within a global team building mass-production autonomous forklifts for warehouses, for KION Group AG.

2023.09 – 2026.01 **Integration and Deployment Engineer**
Center for Research and Technology Hellas (CERTH), Thessaloniki, Greece

2018.09 – 2023.03 **Robotics Research Engineer**
Aristotle University of Thessaloniki, Greece
Electrical and Computer Engineering Department

2016.09 – 2016.11 **Teaching Assistant** · DD2380 Artificial Intelligence
KTH Royal Institute of Technology, Stockholm, Sweden

2011.10 – 2012.03 **Database Designer**
Egnatia Motorway S.A., Thessaloniki, Greece
Design and implementation of a unified database, suitable for the needs of the Instrumental Landslide and Geotechnical Issues monitoring system, in the context of the European Research Program IRIS.

2011.03 – 2011.05 **Database Developer**
Internship · Egnatia Motorway S.A., Thessaloniki, Greece
Design, development and technical and user documentation of a system for data recovery and report-issuing from the company's bridge register using customizable criteria. The application was developed using ORACLE developer tools.

2008.07 – 2009.06 **Telecommunications Engineer**
Hellenic Telecommunications Organisation (OTE S.A.) Thessaloniki, Greece
Remote service in matters of local and wide area networks.


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## Experience in R&D Projects

2023.09 – 2026.01 **RoBétArmé** · *Horizon Europe, GA No. 101058731*
*Center for Research and Technology Hellas (CERTH), Thessaloniki, Greece*

Sole architect and software integrator of the project, deployed at EDF's Hermillon hydroelectric powerplant with Bouygues Construction after two years of continuous integration between the 12 partners of the consortium. Maintainer of project's GitLab repositories. Task planning and Behaviour Tree orchestration for concrete- and metal-additive manufacturing robots. Development of ROS packages for planning, scheduling, orchestrating and triggering ROS Noetic and ROS 2 Humble packages and other software, and their integration with Behaviour Trees. Definition of the development, integration and deployment principles for the 12 software teams of the project and its 50+ dockerised ROS 2 packages. Integration onto three mobile platforms (two ROBOTNIK mobile bases and one trailer). Integration of in-house and third-party ROS Noetic and ROS 2 Humble packages. Work with 3D LIDARs, RGBD cameras, robot URDFs. Translation of custom interfaces between ROS 1 and ROS 2. Network configuration and troubleshooting for ROS 2 via DDS, Zenoh. Virtualisation via Docker and template specification for all project images. Identification of the bottlenecks that cut deployment time by more than 10x across the fleet of the additive manufacturing robots. General software and hardware troubleshooting. Documentation. Authoring of project deliverables.

2020.04 – 2023.03 **CULTUREID**
*Aristotle University of Thessaloniki, Greece*
*Electrical and Computer Engineering Department*

Development of ROS packages for mapping, localisation, navigation, visualisation (Tkinter). Integration onto one mobile platform (Turtlebot) of a full-stack autonomous social robot (NLP, navigation, SLAM, RFID, HRI and GUI), which has been engaging 2,000+ visitors per year at the AMTh museum since 2023. Sole robot-related integrator. Integration of RFID-related factory- and custom-built software and wrapping into ROS. General software and hardware troubleshooting and integration. Work with 2D LIDARs, RGBD cameras, RFID readers. Documentation. Authoring of project deliverables.

2018.09 – 2021.08 **RELIEF**
*Aristotle University of Thessaloniki, Greece*
*Electrical and Computer Engineering Department*

Development of ROS packages for mapping, localisation, navigation, and visualisation (Qt). Integration onto two ground mobile platforms (Turtlebot 2 & Robotnik RB1) and aerial platforms for RFID inventorying. Localisation accuracy of the fleet raised by 50%, down to cm accuracy. Sole robot-related integrator. Integration of RFID-related factory- and custom-built software and wrapping into ROS. General software and hardware troubleshooting and integration. Work with 2D LIDARs, RGBD cameras, RFID readers. Documentation. Authoring of project deliverables.


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## Voluntary Experience

2013.10 – 2014.07 **Computer Vision Engineer** · PANDORA Robotics Undergrad Team, Electrical and Computer Engineering Department, Aristotle University of Thessaloniki

Design of the architecture, implementation and thorough documentation of the Hole Detection system of the PANDORA robot under ROS, using RGB+Depth sensors (Microsoft Kinect and ASUS Xtion) in the context of the conditions of the international RoboCup Rescue competition.

## Open Source Contributions

Linux `gitrl-z`
Git reflog walker for untangling tree mishaps.

ROS 2 `hitch_estimation_apriltag_array`
Estimation of the angle between a vehicle and a hitched trailer.

ROS 2 `pointcloud_conversions`
Format conversions and storage functions for pointclouds.


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## Education

2018.09 – 2023.06 **Doctorate**
Aristotle University of Thessaloniki, Greece
Electrical and Computer Engineering Department

**Thesis** · 2D LIDAR sensor pose estimation via scan–to–map-scan matching
Advisor: Prof. Georgios Sergiadis, Department of Telecommunications
Defended: 28/06/2023
Committee: Georgios Sergiadis (AUTh), Andreas Symeonidis (AUTh), Traianos Yioultsis (AUTh), Zoe Doulgeri (AUTh), Nikolaos Fachantidis (UoM), Aggelos Bletsas (TUC), Anastasios Delopoulos (AUTh)

2015.09 – 2017.06 **Master of Science**
KTH Royal Institute of Technology, Stockholm, Sweden
School of Electrical Engineering
Programme title: *Systems, Control, and Robotics*

**Thesis** · Robust Decentralized Control of Cooperative Multi-robot Systems: An inter-constraint Receding Horizon approach
Advisor: Prof. Dimos Dimarogonas, Department of Automatic Control

2005.09 – 2013.07 **Diploma**
Aristotle University of Thessaloniki, Greece
Electrical and Computer Engineering Department
GPA: 7.94/10.0
Class rank: 23/280 – $92^{nd}$ percentile

**Thesis** · Multi-label classification using Learning Classifier Systems
Advisor: Prof. Pericles Mitkas, Department of Electronics and Computer Engineering
Committee: Pericles Mitkas (AUTh), Anastasios Delopoulos (AUTh), Andreas Symeonidis (AUTh)


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# Publications

Link to Google Scholar

**Alexandros Filotheou**. "CBGL: Fast Monte Carlo Passive Global Localisation of 2D LIDAR Sensor". In: *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2024, pp. 3268–3275. doi: 10.1109/IROS58592.2024.10802235

Anastasios Tzitzis, **Alexandros Filotheou**, Aristidis Raptopoulos Chatzistefanou, Traianos Yioultsis, and Antonis G. Dimitriou. "Real-Time Global Localization of a Mobile Robot by Exploiting RFID Technology". In: *IEEE Journal of Radio Frequency Identification* (2023), pp. 1–1. issn: 2469-7281. doi: 10.1109/JRFID.2023.3288982. url: https://ieeexplore.ieee.org/document/10160120/

**Alexandros Filotheou**, Andreas L. Symeonidis, Georgios D. Sergiadis, and Antonis G. Dimitriou. "Correspondenceless scan-to-map-scan matching of 2D panoramic range scans". In: *Array* 18 (July 2023), p. 100288. issn: 25900056. doi: 10.1016/j.array.2023.100288. url: https://linkinghub.elsevier.com/retrieve/pii/S2590005623000139

**Alexandros Filotheou**, Georgios D. Sergiadis, and Antonis G. Dimitriou. "FSM: Correspondenceless scan-matching of panoramic 2D range scans". In: *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Oct. 2022, pp. 6968–6975. doi: 10.1109/IROS47612.2022.9981228

**Alexandros Filotheou**. "Correspondenceless scan-to-map-scan matching of homoriented 2D scans for mobile robot localisation". In: *Robotics and Autonomous Systems* 149 (Mar. 2022), p. 103957. issn: 09218890. doi: 10.1016/j.robot.2021.103957. url: https://linkinghub.elsevier.com/retrieve/pii/S0921889021002323

**Alexandros Filotheou**, Anastasios Tzitzis, Emmanouil Tsardoulias, Antonis Dimitriou, Andreas Symeonidis, George Sergiadis, and Loukas Petrou. "Passive Global Localisation of Mobile Robot via 2D Fourier-Mellin Invariant Matching". In: *Journal of Intelligent & Robotic Systems* 104.2 (Feb. 2022), p. 26. issn: 0921-0296. doi: 10.1007/s10846-021-01535-7. url: https://link.springer.com/10.1007/s10846-021-01535-7

George Mylonopoulos, Aristidis Raptopoulos Chatzistefanou, **Alexandros Filotheou**, Anastasios Tzitzis, Stavroula Siachalou, and Antonis G. Dimitriou. "Localization, Tracking and Following a Moving Target by an RFID Equipped Robot". In: *2021 IEEE International Conference on RFID Technology and Applications (RFID-TA)*. IEEE, Oct. 2021, pp. 32–35. isbn: 978-1-6654-2657-2. doi: 10.1109/RFID-TA53372.2021.9617436. url: https://ieeexplore.ieee.org/document/9617436/

Antonis Dimitriou, Anastasios Tzitzis, **Alexandros Filotheou**, Spyros Megalou, Stavroula Siachalou, Aristidis R. Chatzistefanou, Andreana Malama, Emmanouil Tsardoulias, Konstantinos Panayiotou, Evaggelos Giannelos, Thodoris Vasiliadis, Ioannis Mouroutsos, Ioannis Karanikas, Loukas Petrou, Andreas Symeonidis, John Sahalos, Traianos Yioultsis, and Aggelos Bletsas. "Autonomous Robots, Drones and Repeaters for Fast, Reliable, Low-Cost RFID Inventorying & Localization". In: *2021 6th International Conference on Smart and Sustainable Technologies (SpliTech)*. IEEE, Sept. 2021, pp. 01–06. isbn: 978-953-290-112-2. doi: 10.23919/SpliTech52315.2021.9566425. url: https://ieeexplore.ieee.org/document/9566425/

**Alexandros Filotheou**, Emmanouil Tsardoulias, Antonis Dimitriou, Andreas Symeonidis, and Loukas Petrou. "Pose Selection and Feedback Methods in Tandem Combinations of Particle Filters with Scan-Matching for 2D Mobile Robot Localisation". In: *Journal of Intelligent & Robotic Systems* 100.3-4 (Dec. 2020), pp. 925–944. issn: 0921-0296. doi: 10.1007/s10846-020-01253-6. url: https://link.springer.com/10.1007/s10846-020-01253-6

Anastasios Tzitzis, Spyros Megalou, Stavroula Siachalou, Tsardoulias G. Emmanouil, **Alexandros Filotheou**, Traianos V. Yioultsis, and Antonis G. Dimitriou. "Trajectory Planning of a Moving Robot Empowers 3D Localization of RFID Tags With a Single Antenna". In: *IEEE Journal of Radio Frequency Identification* 4.4 (Dec. 2020), pp. 283–299. issn: 2469-7281. doi: 10.1109/JRFID.2020.3000332. url: https://ieeexplore.ieee.org/document/9109328/


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Anastasios Tzitzis, **Alexandros Filotheou**, Stavroula Siachalou, Emmanouil Tsardoulias, Spyros Megalou, Aggelos Bletsas, Konstantinos Panayiotou, Andreas Symeonidis, Traianos Yioultsis, and Antonis G. Dimitriou. "Real-time 3D localization of RFID-tagged products by ground robots and drones with commercial off-the-shelf RFID equipment: Challenges and Solutions". In: *2020 IEEE International Conference on RFID (RFID)*. IEEE, Sept. 2020, pp. 1–8. isbn: 978-1-7281-5576-0. doi: 10.1109/RFID49298.2020.9244904. url: https://ieeexplore.ieee.org/document/9244904/

**Alexandros Filotheou**, Emmanouil Tsardoulias, Antonis Dimitriou, Andreas Symeonidis, and Loukas Petrou. "Quantitative and Qualitative Evaluation of ROS-Enabled Local and Global Planners in 2D Static Environments". In: *Journal of Intelligent & Robotic Systems* 98.3-4 (June 2020), pp. 567–601. issn: 0921-0296. doi: 10.1007/s10846-019-01086-y. url: http://link.springer.com/10.1007/s10846-019-01086-y

Anastasios Tzitzis, Spyros Megalou, Stavroula Siachalou, Traianos Yioultsis, Athanasios Kehagias, Emmanouil Tsardoulias, **Alexandros Filotheou**, Andreas Symeonidis, Loukas Petrou, and Antonis G. Dimitriou. "Phase ReLock - Localization of RFID Tags by a Moving Robot". In: *2019 13th European Conference on Antennas and Propagation (EuCAP)*. 2019, pp. 1–5

Spyros Megalou, Anastasios Tzitzis, Stavroula Siachalou, Traianos Yioultsis, John Sahalos, Emmanouil Tsardoulias, **Alexandros Filotheou**, Andreas Symeonidis, Loukas Petrou, Aggelos Bletsas, and Antonis G. Dimitriou. "Fingerprinting Localization of RFID tags with Real-Time Performance-Assessment, using a Moving Robot". In: *2019 13th European Conference on Antennas and Propagation (EuCAP)*. 2019, pp. 1–5

**Alexandros Filotheou**, Alexandros Nikou, and Dimos V. Dimarogonas. "Robust decentralised navigation of multi-agent systems with collision avoidance and connectivity maintenance using model predictive controllers". In: *International Journal of Control* 93.6 (June 2020), pp. 1470–1484. issn: 0020-7179. doi: 10.1080/00207179.2018.1514129. url: https://www.tandfonline.com/doi/full/10.1080/00207179.2018.1514129

**Alexandros Filotheou**, Alexandros Nikou, and Dimos V. Dimarogonas. "Decentralized Control of Uncertain Multi-Agent Systems with Connectivity Maintenance and Collision Avoidance". In: *2018 European Control Conference (ECC)*. IEEE, June 2018, pp. 8–13. isbn: 978-3-9524-2698-2. doi: 10.23919/ECC.2018.8550343. url: https://ieeexplore.ieee.org/document/8550343/


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## Distinctions

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|---|---|
| 2024 | One of eighteen authors of single-authored papers (1,587 total) presented at the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024) |
| 2016 | Teaching Assistant, DD2380 - Artificial Intelligence, under the supervision of Professor Patric Jensfelt, KTH Royal Institute of Technology, Sweden |
| 2015 | $2^{nd}$ place in Autonomy class in RoboCup Rescue as member of PANDORA robotics team |
| 2013 | Ranked $30^{th}$ in graduating class among 224 students who graduated in 2013, ECE, AUTh, Greece |
| 2011 | Top of my class in the course of Database Systems, winter semester 2010 – 2011, AUTh, Greece |
| 2005 | Ranked $21^{st}$ in entering class among 280 students who enrolled in 2005, ECE, AUTh, Greece |

## Computer Skills

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| Languages | `C/C++`, `Python`, `shell`, `MATLAB/Octave` {`PL/`}`SQL`, `Java`, `Assembly` |
| Robotics/OS | `Linux`, `ROS/ROS 2` |
| Tools/Frameworks | `git`, `Docker`, Eigen, Behavior Trees, Gazebo, `CI/CD`, `Qt/Tkinter`, `OpenCV` |
| Sensors/Hardware | Ouster 3D and Hokuyo 2D LiDARs, SICK Visionary-T, Astra, Realsense, ZED2, Impinj R420 |
| Control Techniques | MPC, PID, LQR |
| Graphics | `AutoCAD`, `Gimp` |

## Languages

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|---|---|
| English | Native / Fluent (IELTS 8.5 - C2 Proficiency) |
| Greek | Mother tongue |

## References

Dr. Antonis Dimitriou · Coordinator of the projects for which I worked during my AUTh years
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For a complete list of references, whether they be supervisors or coworkers, visit

https://github.com/li9i/cv/tree/master/references


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